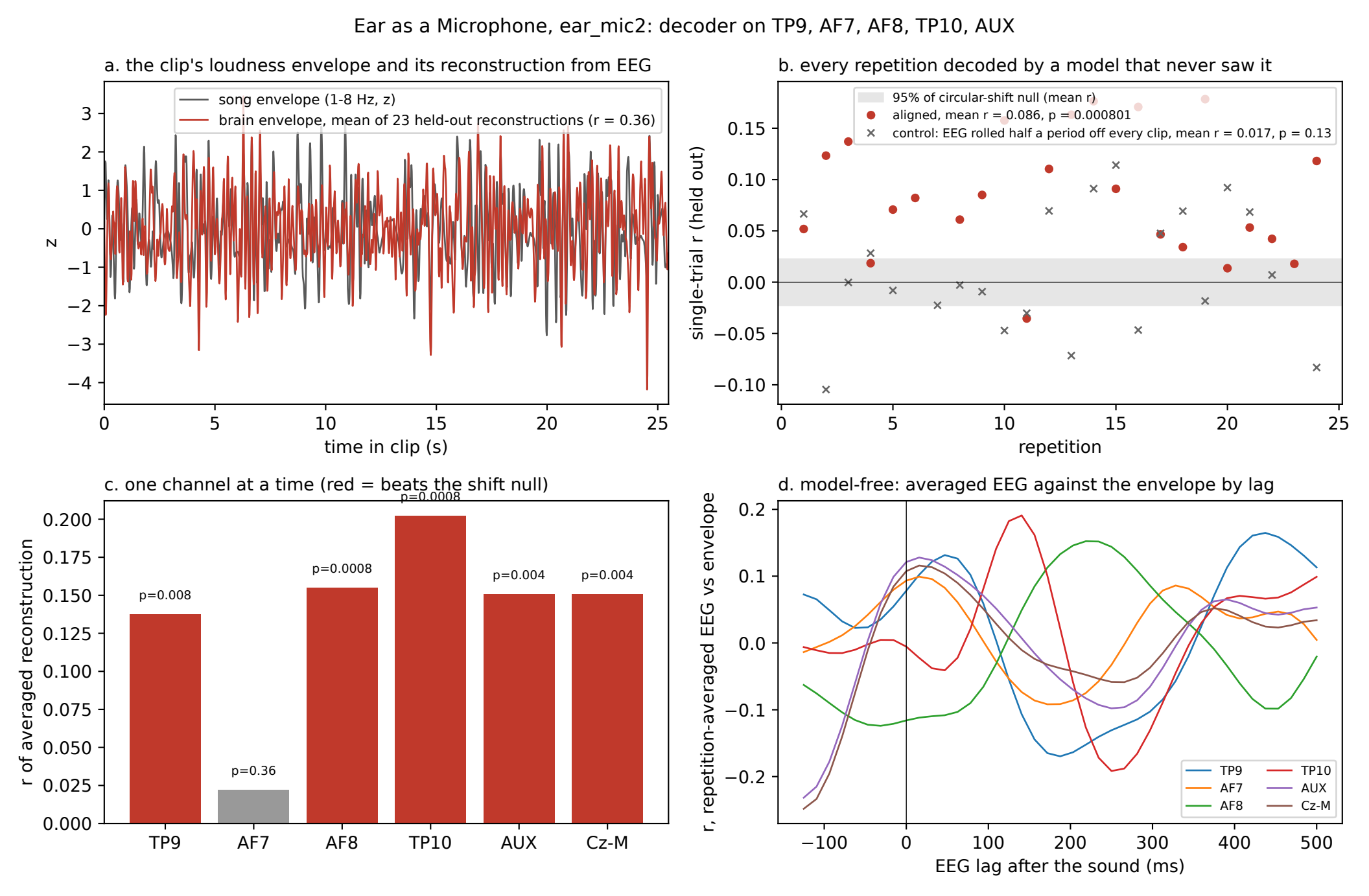
Ear as a Microphone, ear_mic2: decoder on TP9, AF7, AF8, TP10, AUX
a. the clip's loudness envelope and its reconstruction from EEG
song envelope (1-8 Hz, z)
brain envelope, mean of 23 held-out reconstructions (r = 0.36)
z
time in clip (s)
b. every repetition decoded by a model that never saw it
95% of circular-shift null (mean r)
aligned, mean r = 0.086, p = 0.000801
control: EEG rolled half a period off every clip, mean r = 0.017, p = 0.13
single-trial r (held out)
repetition
c. one channel at a time (red = beats the shift null)
r of averaged reconstruction
p=0.008
p=0.36
p=0.0008
p=0.0008
p=0.004
p=0.004
TP9
AF7
AF8
TP10
AUX
Cz-M
d. model-free: averaged EEG against the envelope by lag
r, repetition-averaged EEG vs envelope
EEG lag after the sound (ms)
TP9
AF7
AF8
TP10
AUX
Cz-M